

Job Loss and Marriage Entry: Gender Differences in Responses to Economic Shocks

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Abstract

This paper examines the causal effect of job displacement on marriage and how this effect differs by gender. Using administrative data from South Korea, I exploit job displacement due to plant closures as an exogenous shock. I find that job displacement leads to strikingly different marriage responses for men and women. Displaced men experience a persistent decline in the likelihood of being married, falling by about 1 percentage point on average over the five years following job loss. In contrast, displaced women become more likely to be married, with the increase concentrated in the year of displacement and the following year. Over the same five-year period, the estimated effect on marriage is 1.7 percentage points higher for women than for men. These gender differences are not driven by pre-displacement characteristics; reweighting women to match men's characteristics yields larger estimated effects. Moreover, women's responses are stronger among those experiencing more severe economic disruptions, whereas the patterns for men are less clearly tied to measured labor-market losses. Among those who marry, spouse characteristics change only modestly, with small declines in socioeconomic characteristics. Overall, these findings suggest that the same economic shock has fundamentally different implications for marriage for men and women.

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1 Introduction

There is widespread interest among social scientists in the relationship between economic conditions and marriage decisions. A classic economic framework in this area is the [Becker \(1973\)](#) model, which posits that the gains from marriage arise from the specialization of labor within the household—a process that depends on the relative labor income of partners. This framework implies gender-differentiated effects of economic conditions on marriage formation: a stronger economic position for men increases incentives to marry, whereas a stronger economic position for women may reduce them by raising the opportunity cost of domestic specialization. Consistent with this prediction, a large empirical literature documents a robust positive relationship between men’s economic position and marriage decisions, while the evidence for women is more mixed and varies across institutional and cultural contexts¹. Although prior research documents important associations between economic conditions and marriage decisions, credible evidence on the causal relationship remains limited. This is in part because economic conditions and marriage decisions are jointly determined, making it difficult to distinguish cause from effect in the correlations between them.

In this paper, I study the causal effects of job displacement on marriage and whether these effects differ by gender. I exploit a quasi-experimental setting created by job losses from plant closures among unmarried Korean workers. A large literature has used job displacement from mass layoffs or plant closures as an individual-level exogenous economic shock, as such events separate all or most workers irrespective of individual performance ([Charles and Stephens, 2004](#); [Del Bono et al., 2012](#); [Halla et al., 2020](#); [Huttunen and Kellokumpu, 2016](#); [Illing et al., 2024](#); [Jacobson et al., 1993](#); [Sullivan and von Wachter, 2009](#)). These studies also document that job displacement leads to substantial and persistent earnings losses, as well as declines in job stability, employment rates, and broader economic well-being. This setting therefore allows me to examine whether the causal effect of negative economic shocks on marriage formation differs by gender.

To conduct this analysis, I combine four types of confidential administrative microdata from Korea—the Employer–Employee data, the Business Register, the Population Census, and the Population Dynamic Census—covering the period 2015 to 2023. I convert annual information in these datasets into a quarterly panel that tracks individuals’ employment status, job separations, firm closures, and marriage events with high temporal precision. Given that these administrative data cover the universe of workers in Korea, all firms that

¹See, for example, [Ahn and Mira \(2001\)](#), [Blau et al. \(2000\)](#), [Burgess et al. \(2003\)](#), [Burstein \(2007\)](#), [Freiden \(1974\)](#), [Goldscheider and Waite \(1986\)](#), [Gutiérrez-Domènech \(2008\)](#), [MacDonald and Rindfuss \(1981\)](#), [Oppenheimer et al. \(1997\)](#), [Preston and Richards \(1975\)](#), [Schneider et al. \(2019\)](#), [Sweeney \(2002\)](#), [Watson and McLanahan \(2011\)](#), [Xie et al. \(2003\)](#), [Ono \(2003\)](#).

close during the period, and all marriage registrations for cohorts born between 1983 and 1995, they allow for a comprehensive and precise examination of how individual-level job displacement affects men’s and women’s transitions into marriage.

I compare displaced and non-displaced workers separately by gender using a dynamic difference-in-differences design. To construct comparable control groups, I implement coarsened exact matching (CEM) within gender, matching on pre-displacement characteristics—including age, tenure, industry, establishment size, firm revenue, and quarterly earnings. I then estimate an event-study specification using the matched sample, tracing individuals from four quarters before to twenty quarters after displacement. Given systematic differences in pre-displacement characteristics between men and women, a concern is that observed gender differences in post-displacement marriage responses reflect compositional differences. To address this concern, I additionally implement a reweighting procedure in the spirit of [Illing et al. \(2024\)](#) that aligns the covariate distribution of women with that of men. This approach serves as a complementary exercise to assess whether the observed gender differences in post-displacement marriage responses persist after accounting for differences in pre-displacement characteristics. I examine heterogeneity to assess whether marriage responses are stronger among individuals facing greater economic instability. In addition, I analyze spousal outcomes among those who marry, allowing me to assess whether job displacement affects not only the likelihood of marriage but also partner characteristics.

The results show that job displacement has sharply different effects on marriage by gender. I first document that displacement generates substantial labor market losses for both men and women. Employment, tenure, and earnings decline sharply after job loss and recover only partially over the following years. Despite these broadly similar economic losses, marriage responses move in opposite directions. For men, job displacement reduces the probability of being married, with the decline becoming persistent after displacement and reaching 1.47 percentage points four years after job loss. In contrast, women become more likely to marry following displacement: the probability of being married increases by 0.99 percentage points in the year of displacement and remains elevated in the following year. On average over the five years after displacement, job loss reduces men’s probability of being married by 1 percentage point, while its estimated effect is 1.7 percentage points higher for women than for men. When women are reweighted to match men’s pre-displacement characteristics, the estimated gender difference in the effect increases to 2.4 percentage points. This pattern is consistent with the fact that reweighted women experience more severe economic disruption, suggesting that the stronger marriage response among women is not driven by compositional differences. In sum, these results indicate that men and women respond differently to negative economic shocks, rather than reflecting differences in pre-displacement characteristics. Heterogeneity

analyses further show that women’s marriage responses are concentrated among those facing more severe or persistent economic disruption, such as those who fail to become re-employed, lower-earning workers, and older women. For men, however, the subgroup patterns are less clear: marriage declines are larger among those who do not become re-employed, but do not vary consistently with pre-displacement earnings and are more pronounced among younger men. These results suggest that men’s marriage responses are not fully explained by measured labor market loss severity. Finally, among those who marry after displacement, I find limited but suggestive evidence that job loss is associated with changes in spouse characteristics along socioeconomic dimensions. Displaced workers are less likely to marry college-educated spouses and tend to marry spouses with slightly lower pre-marriage earnings. However, there is little evidence of broad changes in other observable spouse characteristics, such as spousal age or labor-force attachment.

This paper contributes to the literature on economic conditions and marriage in several ways. First, it provides direct evidence on how economic shocks affect marriage differently by separately examining men’s and women’s marriage outcomes, using job displacement due to plant closures as a plausibly exogenous shock. Given the difficulty of isolating the causal relationship between economic conditions and marriage, recent work has exploited exogenous labor demand shocks to identify causal effects on marriage outcomes, including [Kearney and Wilson \(2018\)](#), [Autor et al. \(2019\)](#), and [Ragab and Saad \(2023\)](#). While these studies provide important causal evidence linking economic conditions to marriage, their primary research question centers on the role of men’s economic conditions and the concept of “marriageable men”, even when both men and women are included in the analysis. For example, [Kearney and Wilson \(2018\)](#) examine whether improvements in male economic opportunities, driven by positive labor demand shocks, lead to changes in marriage and fertility. They show that such shocks primarily increase fertility, with limited effects on marriage in recent periods, suggesting that improvements in economic conditions are more likely to translate into nonmarital childbearing rather than marriage. [Autor et al. \(2019\)](#) exploit gender-specific labor demand shocks to study how changes in the relative economic conditions of men and women affect marriage. Using women’s marital status as the primary outcome, they find that negative shocks to male-intensive industries reduce marriage rates, which they interpret as reflecting a decline in the supply of marriageable men. Similarly, [Ragab and Saad \(2023\)](#) document that a large negative economic shock reduces marriage rates among young men. Taken together, this literature highlights the importance of economic conditions for marriage. However, it largely centers on men’s economic conditions, even when women are included in the analysis. In contrast, this paper treats both men’s and women’s

responses as central research questions and examines how each responds to the same economic shock within a unified empirical framework.

This paper also complements the literature that exploits local labor demand shocks by focusing on an individual-level economic shock. Existing studies have provided causal evidence by examining regional or sectoral economic shocks. Such shocks affect not only individuals' economic conditions but also broader features of the local marriage market, including the pool of potential partners and overall economic conditions. As a result, these approaches capture both individual responses and broader equilibrium effects, making it difficult to separate the impact of a worker's own economic disruption from changes in the surrounding environment. By focusing on individual-level job displacement, this paper provides a more direct assessment of the role of a worker's own economic shock in shaping marriage outcomes. This approach offers a complementary perspective to the existing literature.

In addition, this paper contributes to the job displacement literature by examining a largely unexplored margin: how job loss affects entry into marriage among individuals who are initially unmarried. One recent study examines remarriage and marital sorting after displacement, focusing on previously partnered men rather than individuals who are initially unmarried (Foerster et al., 2024). Nevertheless, most prior research on the family consequences of job displacement has primarily examined outcomes such as divorce and fertility. Prior work shows that job displacement can significantly affect marital stability. As Charles and Stephens (2004) show, job loss—particularly among husbands—increases the risk of marital dissolution, suggesting that adverse economic shocks can alter the gains from marriage. Consistent with this, subsequent studies document that job displacement raises the likelihood of divorce and family dissolution (Doiron and Mendolia, 2012; Keldenich and Luecke, 2020), although the effects may differ between husbands' and wives' job loss (Eliason, 2012). Job loss has also been shown to influence fertility decisions, with evidence that negative income shocks reduce childbearing. For instance, Del Bono et al. (2012) and Huttunen and Kellokumpu (2016) document that job loss is associated with reductions in fertility, particularly among women with stronger labor market attachment. Despite this evidence on post-marriage outcomes, little attention has been paid to marriage entry itself. Yet marriage entry represents a fundamental margin of household formation and precedes many subsequent family outcomes. South Korea provides a useful setting for studying this margin because marriage remains the primary institutional context for childbearing and subsequent family outcomes, while nonmarital childbearing and cohabitation remain relatively uncommon. As Stevenson and Wolfers (2007) document, family behavior is inherently dynamic, with marriage, divorce, and fertility evolving over time as part of an interconnected life cycle. Examining any one of these outcomes in isolation may therefore yield an incomplete picture of how labor market

conditions shape family behavior. By identifying the effect of job loss on marriage entry, this study helps fill an important gap and provides a more comprehensive understanding of the family consequences of job displacement.

The paper proceeds as follows. Section 2 describes the data and explains how the treatment and control groups are constructed. Section 3 outlines the empirical strategy. Section 4 presents the main results on the impact of job displacement on labor market and marital outcomes of unmarried young adults. Section 5 concludes.

2 Data

2.1 Korean Administrative Data

This study relies on four confidential administrative datasets from Korea: the Employer–Employee Data, the Business Register, the Population Dynamic Census, and the Population Census—covering the period 2015 to 2023. Together, these datasets cover the universe of establishments and workers in Korea and allow me to track individuals’ labor market histories, firm closures, and marriage events with high precision.

The Employer–Employee Data is an annual panel dataset that includes all job information for the entire population each year. These records are drawn from Korea’s administrative systems, including the four major social insurances and the National Tax Service. One key feature of this dataset is that employers are identified not at the establishment level but at the business-group level, meaning that all establishments belonging to the same corporate group share a common employer identifier. While demographic information is limited to age and sex, the dataset contains rich employment details, including hire and separation dates², annual earnings, employment type, and basic business-group characteristics such as industry and indicators for employment-size categories.

The Business Register is an annual census of all domestic establishments in Korea, constructed from a combination of administrative records—such as business registration files—and statistical data sources. It includes information on establishment openings and closures, industry, revenues, and other establishment-level characteristics. Because the Employer–Employee Data do not contain information on whether or when an employer closes, I combine them with the Business Register to identify the precise date of each establishment’s closure and to supplement worker records with additional firm characteristics (e.g., revenues).

²The hire date and end date are recorded in year-month format. And if a worker is employed until the end of the year, the end date for that year is recorded as December of the same year.

The Population Dynamic Census includes information on individuals born (or registered) in Korea between 1983 and 1995 as of 2023, covering dates of birth, marriage, divorce, and death, as well as individuals' education, age, and other event-specific characteristics. In addition, for marriage, birth, and divorce events, the dataset provides parallel information for spouses, including their age, education, and other characteristics measured at the corresponding event date. I merge the Population Dynamic Census with the Employer–Employee–Business Register linked data at the worker level to identify the exact timing of each worker's marriage and to obtain the corresponding characteristics of the spouse.

The Population Census is an annual, registry-based dataset constructed from administrative sources, including the resident registration system, foreign resident registration records, and the family relations registry. It covers the universe of domestic and foreign residents living in Korea as of November 1 of each year. Since the Population Census provides worker identifiers and relationship-to-household-head codes, I am able to identify the spouse of each married worker if the couple is living together in that year. The spouse identifier obtained from the Population Census allows me to match each married worker to the spouse's employment and earnings records in the Employer–Employee Data. In combination with the spouse information already recorded in the Population Dynamic Census, this linkage enables a comprehensive comparison of spouse characteristics—including earnings, education, and age—between the treatment and control groups, both of whom married after the closure event.

As outlined above, the analysis relies on an annual panel dataset constructed by linking the four administrative datasets described in this section. Although the full dataset spans 2015 to 2023, employer identifiers in the Employer–Employee Data are available only from 2015 to 2020. As a result, the identification of job displacement due to firm closure is limited to this period. In addition, to avoid potential confounding effects of the COVID-19 pandemic, the analysis focuses on plant closures occurring between 2016 and 2018. This restriction ensures that a meaningful portion of the post-displacement period remains unaffected by pandemic-related shocks. In the empirical analysis, the data are further transformed into a quarterly panel to leverage the timing of key events. This is possible because key date variables—such as hire dates, separation dates, marriage dates, and plant-closing dates—are recorded in year–month format and can be mapped into calendar quarters.³ I therefore construct a quarterly panel spanning from 2015Q1 to 2023Q4, which provides a longer effective time series for the empirical analysis.

³Because these date variables are provided in year–month format, quarterly employment, marriage, and plant-closing statuses can be identified by assigning each year–month value to its corresponding quarter. This allows an annual panel dataset to be transformed into a quarterly one.

In the analysis, I restrict the sample to workers who were unmarried in the quarter immediately preceding displacement. These displaced workers are matched to a control group of single individuals who did not experience job loss during the same period. The final treated sample consists of 18,701 displaced workers—10,165 women and 8,536 men. Because each displaced worker is matched to one control individual, the matched analysis sample contains 37,402 individuals in total. The analysis is restricted to individuals born between 1983 and 1995, corresponding to the coverage of the Population Dynamic Census, which provides marriage-date information only for these cohorts. Given that the average age at first marriage in recent years is approximately 33–34 for men and 30–31 for women in South Korea, this restriction ensures that the sample remains within the age range for which marriage entry is empirically relevant. Information on all sample members is available from 2015Q1 to 2023Q4, and the empirical analysis focuses on plant closures occurring between 2016Q1 and 2018Q4.

2.2 Definition of Job Displacement

A large body of existing literature that analyzes the effects of job displacement defines a worker as displaced if the worker leaves their stable job at their main employer during a mass layoff or plant closure (Charles and Stephens, 2004; Davis and von Wachter, 2011; Del Bono et al., 2012; Halla et al., 2020; Huttunen and Kellokumpu, 2016; Illing et al., 2024; Jacobson et al., 1993; Schmieder et al., 2023; Sullivan and von Wachter, 2009). This definition focuses on workers who likely lost their jobs involuntarily. However, due to data limitations—specifically, the inability to identify mass layoffs—I define displacement exclusively as job loss resulting from a plant closure. This restriction is unlikely to compromise the validity of the analysis, as plant closures represent a source of job loss that is largely unrelated to workers’ own characteristics and therefore more plausibly exogenous than mass layoffs. In contrast, although mass layoffs involve sizable separations within a firm, the selection of which workers are laid off is often non-random, making displacement potentially correlated with worker performance or employer-specific assessments.

It is important to note that, to ensure accurate identification of displacement events, the definition of plant closure is further restricted to business groups that operate a single establishment. This restriction is required because, given the structure of the administrative data, establishment-level closure information can be reliably linked to workers’ employment records only in such cases.⁴ While this restriction improves accuracy in identifying truly

⁴The Employer–Employee Data provide employer identifiers only at the business-group level, whereas the Business Register includes both business-group and establishment identifiers, along with establishment-level closure dates. Therefore, it is possible to determine whether a worker was employed at a closed establishment only when the business group operates a single establishment. For multi-establishment business groups,

exogenous job losses, it inevitably narrows the set of closures analyzed relative to settings where establishment-level closures can be directly identified. As a result, the analysis is limited to workers employed at relatively small, single-establishment firms.⁵

More specifically, I define a worker as displaced in quarter $t = c$ if their main employment⁶ at the establishment ends either on the recorded closure date or within the three months preceding it. I further restrict the sample to workers who experience their first observed plant closure during the observed period to ensure a clean treatment definition. To capture economically meaningful closures, I define plant closure based on specific criteria recorded in the Business Register, including only closures attributed to business downturns, administrative dispositions, or comparable cessation reasons. Cases resulting from mergers, takeovers, seasonal shutdowns, or other non-terminal events are excluded to avoid misclassification and to focus on the precise identification of true closures. In addition, I restrict the sample to workers with at least one year of tenure at the establishment in quarter $t = c - 1$, who are employed as regular workers and work at establishments with at least five employees. These restrictions are intended to exclude workers with relatively weak attachment to their employers. While prior studies often impose higher tenure thresholds—2–3 years—and firm-size thresholds—30–50 employees—these requirements are relaxed here. The tenure requirement is lowered given the focus on younger workers relative to the existing job displacement literature, while the firm-size threshold is reduced to maintain sufficient sample size under the restriction to single-establishment firms.⁷ Finally, the analysis focuses on plant closures occurring between 2016Q1 and 2018Q4, allowing each worker to be tracked from four quarters before to twenty quarters after displacement.

2.3 Constructing Treatment and Control Group

The construction of the analysis sample proceeds in two steps. First, I identify all workers who satisfy the baseline sample restrictions. Second, I use coarsened exact matching (CEM) to construct a comparable control group for the displaced workers.

This study focuses on workers who satisfy the following baseline restrictions. Workers must be born between 1983 and 1995; be employed at a single-establishment firm with at

the Employer–Employee Data do not indicate the specific establishment at which a worker was employed, making it impossible to determine whether the worker was affected by an establishment closure.

⁵See Appendix Table B.1, which compares the characteristics of single- and multi-establishment firms using the 2016 Employer–Employee Data and Business Register.

⁶Main employment is defined as the job with the longest ex-post tenure. This definition is intended to capture a relatively more stable employment relationship within the quarter.

⁷Imposing stricter restrictions—requiring at least two years of tenure and employment at establishments with at least 50 employees—reduces the sample by 86%. Nevertheless, the results are qualitatively similar, although long-run effects become somewhat attenuated.

least five employees at quarter $t = c - 4$ (i.e., one year prior to the event date); have at least one year of tenure and hold a regular-worker position at $t = c - 1$; and be unmarried at $t = c - 1$.⁸ Since the treatment and control groups may have different underlying marriage tendencies, I further examine a specification that restricts the sample to individuals who are unmarried at $t = c - 4$, while allowing transitions into marriage between $t = c - 3$ and $t = c - 1$. This specification relaxes the baseline restriction and enables a test for differential pre-treatment marriage dynamics across groups. The results indicate that the marriage trends of the two groups are parallel up to two quarters before displacement, but begin to diverge one quarter prior to job loss. This pattern suggests that the identifying assumption of parallel trends is reasonable over the relevant pre-treatment period and that the main results are unlikely to be driven by differential pre-treatment marriage dynamics.

To construct an appropriate control group of non-displaced workers, I apply coarsened exact matching (CEM) without replacement and match each displaced worker to one control worker. The matching is implemented separately for men and women. Matching is performed on age at $t = c - 1$, establishment size at $t = c - 4$, industry at $t = c - 4$, tenure at $t = c - 1$, and quantiles of revenue and quarterly total earnings at $t = c - 4$ (see Appendix A. for details).

Table 1 reports descriptive statistics for displaced and matched non-displaced workers separately by gender. Panel A reports individual-level characteristics, and Panel B reports establishment-level characteristics, all measured prior to displacement. As shown in the table, the sample consists of relatively young workers, with an average age of 26 for women and 28 for men. In addition, because the analysis focuses on displacement from single-establishment firms, the sample is disproportionately composed of workers employed at smaller establishments, with the share declining as establishment size increases. Following Austin (2009), covariate balance is assessed using standardized mean differences (SMDs), which provide a scale-invariant measure of differences between groups and are widely used regardless of the specific matching procedure. SMD is defined as the difference in means between the two groups divided by the pooled standard deviation, with values below 0.1 in absolute terms commonly considered indicative of acceptable covariate balance. After matching, the SMDs are close to zero across all covariates for both the female and male samples, indicating that the matched control group closely resembles the displaced group in observed pre-displacement characteristics. This is further confirmed in Appendix B.1, which plots SMDs before and after matching for each covariate. While the raw (unmatched)

⁸The baseline is defined as the quarter immediately preceding the event date, $t = c - 1$. However, because firm-level characteristics (e.g., revenue, employment size, industry) are available only annually in the administrative data, the corresponding baseline information is taken from $t = c - 4$, which represents one year prior to the event.

sample exhibits meaningful imbalances in several variables—particularly quarterly earnings and revenue—matching substantially reduces these differences, bringing all SMDs well within the ± 0.1 threshold. Overall, the matching procedure successfully constructs comparable treatment and control groups for both men and women, lending credibility to the subsequent difference-in-differences estimates.

3 Empirical Strategy

3.1 Event Study

To estimate the impact of job displacement on marriage and labor market outcomes, I implement an event-study design. This approach traces the evolution of outcomes before and after displacement, thereby identifying the dynamic effects of the shock. The model is estimated separately for men and women:

$$Y_{itc} = \sum_{l=-4}^{20} \gamma_l \times I[t = c + l] + \sum_{l=-4}^{20} \delta_l \times I[t = c + l] \times D_i + \pi_t + \alpha_i + \epsilon_{itc} \quad (1)$$

where Y_{itc} denotes the outcome for individual i observed in calendar quarter t , with event time l normalized so that zero corresponds to the quarter of displacement. D_i is an indicator equal to one if individual i is displaced. $I[t = c + l]$ denotes an event-time indicator that equals one in quarter t when the period is l quarters relative to the displacement quarter c . The quarter immediately preceding displacement ($l = -1$) is omitted as the reference period. The coefficients δ_l measure how outcomes for displaced workers evolve relative to those of comparable non-displaced workers at each event time l . π_t and α_i denote calendar-quarter and individual fixed effects, respectively, and ϵ_{itc} is the error term. Standard errors are clustered at the company level, where matched control workers are assigned to the same displacement event as their treated counterparts.

The outcome set includes the main outcome of interest and several labor market outcomes. The main outcome is an indicator for being married, which equals one if the individual has ever entered marriage by quarter t and zero otherwise. The labor market outcomes are: (i) *Any labor income*, an indicator equal to one if the individual holds any job with positive earnings in quarter t , regardless of employment type; (ii) *regular employment*, an indicator equal to one if the individual holds a job with a contract duration exceeding one month in quarter t ; (iii) *longest tenure*, measured as the number of months of tenure in the job with the longest tenure spell held during quarter t ; and (iv) *quarterly earnings*, defined as total earnings received in quarter t , excluding earnings from special-type and freelance employment.

3.2 Aggregate Difference-in-Differences

To provide a more concise summary of treatment effects, I also estimate a difference-in-differences specification that aggregates event-time indicators into annual periods relative to displacement.

$$\begin{aligned}
Y_{itc} = & \beta_0 (\mathbf{1}\{\text{During}_t\} \times D_i) + \sum_{s=1}^4 \beta_s (\mathbf{1}\{\text{After}_{s,t}\} \times D_i) \\
& + \gamma_0 \mathbf{1}\{\text{During}_t\} + \sum_{s=1}^4 \gamma_s \mathbf{1}\{\text{After}_{s,t}\} + \pi_t + \alpha_i + \epsilon_{itc}
\end{aligned} \tag{2}$$

where $\mathbf{1}\{\text{During}_t\}$ indicates the year of displacement, and $\mathbf{1}\{\text{After}_{s,t}\}$ denotes indicators for the s -th year after displacement. The omitted category is the year prior to displacement. The coefficients β_0 and β_s capture the average treatment effects in the displacement year and subsequent post-displacement years, respectively. γ_0 and γ_s absorb common time effects. π_t and α_i denote calendar-quarter and individual fixed effects, respectively.

3.3 Adjusting for Gender Differences: Reweighting Procedure

As shown in Table 1, even under identical baseline restrictions, men and women differ systematically in their pre-displacement characteristics, including earnings, industry, and establishment attributes. This raises the concern that the observed gender differences in post-displacement marriage outcomes may reflect differences in pre-displacement characteristics rather than true gender-specific responses.

To address this concern, I implement a reweighting procedure that aligns the distribution of pre-displacement characteristics of women with that of men, following the approach of DiNardo et al. (1996) and Illing et al. (2024). This exercise serves as a complementary benchmark to assess whether the estimated gender differences persist after accounting for differences in initial conditions. Conceptually, this adjustment evaluates women’s outcomes as if they had the same distribution of pre-displacement characteristics as men, thereby isolating the role of compositional differences.

I estimate a probit model for displaced workers with a male indicator as the dependent variable and use the predicted probabilities to construct the reweighting function $\phi(x)$ that aligns women’s covariate distribution with that of men. The specification includes the same individual and establishment characteristics used in the coarsened exact matching—age($c-1$), total quarterly earnings($c-4$), industry($c-4$), establishment size($c-4$), and tenure($c-1$).

From this regression, I obtain the predicted propensity score $\hat{p}$ and construct the reweighting function as $\phi(x) = \hat{p}/(1 - \hat{p})$. This function assigns larger weights to women whose pre-displacement characteristics more closely resemble those of displaced men, thereby align-

ing the female covariate distribution with that of men. Table 2 reports descriptive statistics for displaced women and men at the reference quarter. Baseline displaced women differ substantially from displaced men in age, earnings, and establishment characteristics (Columns 1 and 3). After reweighting, women’s covariate distribution more closely resembles that of men (Column 2), with higher earnings and employment in larger and higher-revenue establishments. I apply these weights in the event-study analysis in Section 3.1.

4 Results

4.1 Labor Market Effects of Job Displacement

Before turning to the main results on marriage entry, I begin by examining the labor market effects of job displacement. These results provide important context for interpreting the marriage findings and establish that displacement generates economically meaningful shocks for both men and women in the sample. Figure 1 presents event-study estimates for men and women, along with reweighted estimates for women. Across all outcomes, job displacement leads to large and immediate declines, followed by only partial recovery over the subsequent twenty quarters.

Panel (a) shows the effect on any labor income.⁹ For men, the probability of having any labor income drops sharply, falling by approximately 14.3 percentage points in the year of displacement, and recovers gradually but incompletely over the post-displacement period.¹⁰ Women exhibit a similar pattern, with an initial decline of approximately 15.9 percentage points (Table 3, Column 2). From the year following displacement, recovery is slightly faster for women than for men, although the difference remains modest. As a result, the average post-displacement effects do not differ significantly between the two groups (Table 4, Column 2). Panel (b) reports estimates for regular employment and closely parallels Panel (a), although the overall losses are somewhat larger and recovery is slower.

Panel (c) reports the effects on longest tenure. For men, longest tenure declines sharply following displacement, falling by approximately 22.94 months in the year of displacement. Women exhibit a similar pattern, with an initial decline of about 23.88 months (Table 3, Column 4). In both cases, tenure recovers gradually over time but remains substantially below pre-displacement levels even in the long run. Over the five years following displacement, job

⁹Due to the one-year tenure requirement, employment is mechanically at its maximum in the four quarters prior to the event date.

¹⁰Because employment is measured at the quarterly level, individuals who are displaced during quarter $l = 0$ are still recorded as employed in that quarter. As a result, the decline in employment is first observed in the subsequent quarter ($l = 1$) in Figure 1.

displacement reduces longest tenure by about 19.10 months for men and 18.75 months for women on average (Table 4, Column 4). While the initial decline is slightly larger for women, the gender gap narrows over time.

Panel (d) shows the effects on quarterly earnings. Earnings drop sharply for both men and women in the year of displacement. Specifically, earnings fall by approximately 1.20 million KRW for men and 1.18 million KRW for women during the year of displacement (Table 3, Column 5). Earnings then recover gradually over time but remain below pre-displacement levels even in later periods, consistent with other labor market outcomes. Over the five years following displacement, job displacement reduces quarterly earnings by about 0.68 million KRW for men and 0.46 million KRW for women on average (Table 4, Column 5). Relative to pre-displacement baseline earnings, this corresponds to declines of approximately 9.0% for men and 8.1% for women, indicating that the relative magnitude of earnings losses is broadly similar across genders despite differences in levels.

Overall, losses are largest immediately following displacement and recover only partially over time. Even in the long run, outcomes remain below pre-displacement levels, with little difference between men and women on average, although losses are slightly smaller for women in some outcomes (Table 4, Panel A).

However, once women are reweighted to match the pre-displacement characteristics of men, the estimated losses become somewhat larger and more persistent for women than for men, as shown in Figure 1 and Table 4 (Panel B). This suggests that differences in worker composition may partly mask underlying gender gaps in labor market losses.

These findings show that job displacement constitutes a negative economic shock for both men and women, while raising the concern that simple comparisons of marriage responses between the two groups may be driven by differences in pre-displacement characteristics. However, as shown in Section 4.2, the marriage results not only address this concern but also reinforce the interpretation of the results. In particular, the gender differences in marriage responses are unlikely to be driven solely by differences in initial conditions.

4.2 The Main Effect of Job Displacement on Marriage Entry

Given the substantial labor market effects of job displacement for both men and women, I next examine its impact on marriage outcomes. Before turning to the event-study estimates, Appendix Figure B.2 plots raw mean trends in marriage rates for displaced and matched non-displaced workers, separately by gender. After displacement, displaced men marry at lower rates than comparable non-displaced men, whereas displaced women marry at modestly

higher rates than comparable non-displaced women.¹¹ Figure 2 presents corresponding event-study estimates of the effects of job displacement on the probability of being married. For men, marriage declines gradually following displacement, with the effect becoming persistent from approximately three quarters after job loss. As shown in Table 3 (Panel A, Column 1), the probability of being married decreases by 0.76 percentage points one year after displacement relative to the pre-displacement period, and the decline reaches 1.47 percentage points four years after displacement. In contrast, women exhibit a markedly different pattern. Marriage increases immediately following displacement and remains at an elevated level thereafter. Specifically, the probability of being married increases by 0.99 percentage points in the year of displacement and by 0.94 percentage points in the following year (Table 3, Panel B, Column 1). Over the five years after displacement, men experience an average decline of 1 percentage point in marriage, while its estimated effect is 1.7 percentage points higher for women than for men (Table 4, Column 1).

Taken together, men and women exhibit markedly different marriage responses following job displacement. A natural concern is that these differences may reflect variation in pre-displacement characteristics rather than true behavioral responses. As shown in Section 4.1, a similar pattern arises for labor market outcomes: while raw comparisons suggest little gender difference in losses, reweighting women to match the characteristics of men reveals larger losses for women. To address this concern, I apply the same reweighting approach to the marriage outcomes. Figure 2 shows that reweighted women display a pattern similar to the unweighted estimates for women, with an initial increase in the probability of being married that remains slightly elevated over time. Consistent with this, Table 4 (Panel B) shows that, after reweighting, the estimated effect on marriage is 2.4 percentage points higher for women than for men, compared with 1.7 percentage points in the baseline estimates.

The gender-specific marriage patterns are also robust to imposing stricter baseline sample¹² and to using alternative sets of matching covariates¹³ in both the baseline and reweighted specifications (Appendix Figures B.4 and B.5).

Beyond compositional differences, another concern is that these results may reflect differential marriage trends between the treated and control groups rather than the causal effect of job displacement. To address this, I restrict the sample to individuals who are unmarried up to four quarters prior to displacement and allow for marriage only from three

¹¹Marriage rates are mechanically zero in the displayed pre-displacement period because the sample is restricted to individuals who are unmarried immediately before displacement.

¹²The baseline sample restricts workers to establishments with at least five employees and requires at least one year of tenure. The stricter sample instead requires at least two years of tenure and employment at establishments with at least 50 employees.

¹³The alternative matching specifications exclude revenue variables from the matching covariates and restrict the sample to workers who are displaced only once and controls who are never displaced.

quarters before displacement onward. As shown in Appendix B.3, the marriage trends of the treated and control groups are closely aligned up to two quarters before displacement, supporting the parallel trends assumption over the relevant pre-treatment period. While the two groups begin to diverge slightly in the quarter immediately preceding job loss, this pattern is limited and does not appear to drive the main results.

Overall, the results point to substantial gender differences in marriage responses to job displacement. These patterns are consistent with the framework of Becker (1973), in which the gains from household formation depend on gender-based specialization. As shown in Section 4.1, job displacement leads to substantial declines in economic outcomes. For women, such declines may lower the opportunity cost of home production and increase the gains from marriage. For men, the same shock may reduce the economic resources that contribute to the gains from specialization within the household and may also lower their attractiveness in the marriage market, consistent with the notion of “marriageable men” discussed in Wilson (1987). The reweighting results further suggest that these gender differences are unlikely to be driven solely by differences in pre-displacement characteristics.

4.3 Heterogeneous Effects

I next examine heterogeneity in marriage responses within each gender to further interpret the main results. If women’s increased marriage entry reflects greater gains from marriage following adverse economic shocks, as implied by a Beckerian framework, the response should be stronger among women who experience more severe or persistent labor market disruptions. For men, the same framework predicts the opposite pattern: larger economic shocks should lead to larger declines in marriage entry. In addition to the severity of the economic shock, marriage responses may also depend on individuals’ position in the marriage life cycle, since the scope for adjusting marriage timing may vary with age. Figures 3-5 examine these patterns by comparing marriage responses across re-employment status, pre-displacement earnings, and age.

First, Figure 3 examines heterogeneity by re-employment status, defined as whether an individual obtains regular employment within two quarters after displacement. Re-employment provides a natural proxy for the persistence of the economic shock, since individuals who quickly find new employment are able to offset part of the disruption. In line with the discussion above, marriage responses are concentrated among individuals who do not become re-employed. For this group, employment losses are substantially larger and more persistent. The corresponding results for relative earnings, reported in Appendix Figure B.6, show a similar pattern. Women who do not become re-employed exhibit a larger increase in the

probability of being married, while men in the same group experience a larger decline. By contrast, individuals who quickly return to regular employment show little change in marriage outcomes. These results suggest that marriage responses are stronger when recovery from job displacement is limited.

Figure 4 examines heterogeneity by pre-displacement earnings, comparing individuals in the bottom and top halves of the earnings distribution. Pre-displacement earnings provide another measure of workers' economic vulnerability to job loss, as lower-earning workers may have fewer resources to buffer the shock. For both men and women, higher-earning individuals experience smaller employment losses following displacement. The corresponding relative-earnings estimates in Appendix Figure B.7 show some evidence of larger short-run losses among lower-earning women, while differences across earnings groups are less pronounced for men. Consistent with this pattern, women in the lower-earning group exhibit a somewhat larger and more persistent increase in the probability of being married. For men, the decline in marriage is initially somewhat larger among higher-earning workers, but the two earnings groups exhibit broadly similar declines after the first few quarters. Taken together, these patterns suggest that women's marriage responses are more closely related to economic vulnerability and the severity of displacement losses. For men, however, the subgroup patterns provide weaker support for this interpretation. Marriage responses do not vary systematically with pre-displacement earnings, even though lower-earning men experience larger employment losses. One possible reason is that relative earnings losses do not differ sharply across earnings groups, as shown in Appendix Figure B.7. Still, these results suggest that men's marriage responses are less clearly explained by a simple severity-based interpretation of displacement losses.

Finally, Figure 5 turns to differences by age, comparing individuals under age 30 with those aged 30 and above. Age captures an additional dimension of heterogeneity because it reflects individuals' position in the marriage life cycle and may also be correlated with the extent of post-displacement labor market losses. Women, in particular, may face stronger timing constraints because marriage and fertility decisions are closely linked, which can make the timing of marriage more sensitive to economic shocks. For women, the increase in the probability of being married is larger among those aged 30 and above, with a more persistent rise following displacement. Older women also experience somewhat larger and more persistent declines in employment, and the corresponding relative-earnings estimates in Appendix Figure B.8 show a similar pattern. These patterns are consistent with the interpretation that economic shocks may affect women's marriage timing more strongly when labor market losses coincide with stronger life-cycle timing constraints. For men, the pattern is different. The decline in the probability of being married is more pronounced among those

below age 30, while responses among older men are relatively muted. This age pattern does not appear to be driven solely by larger labor market losses among younger men, suggesting that labor market loss severity alone does not account for the male response. One possible interpretation is that younger men are at an earlier stage of relationship formation, making marriage entry more sensitive to negative economic shocks. In contrast, older men may be more likely to have already formed stable relationships or marriage plans, which could buffer the effect of job displacement on observed marriage outcomes.

In sum, the heterogeneity results provide stronger support for a severity-based interpretation among women than among men. For women, marriage increases are larger among groups facing more severe or persistent labor market disruptions, including those who do not become re-employed, lower-earning women, and women aged 30 and above. These patterns are broadly consistent with the Beckerian interpretation discussed above: adverse economic shocks may increase women’s gains from marriage, especially when weakened labor market prospects coincide with stronger timing constraints related to marriage and fertility. They are also consistent with the view that marriage can provide economic insurance or risk sharing within the household ([Weiss, 1997](#)), although the results should be interpreted as suggestive rather than direct evidence of this mechanism. For men, however, the evidence is less consistent with a simple severity-based interpretation. Marriage declines are larger among men who do not become re-employed, but they do not vary systematically by pre-displacement earnings and are more pronounced among younger men. This suggests that men’s responses may depend not only on the magnitude of economic losses, but also on non-economic factors that shape marriage formation.

4.4 Spousal Characteristics

To further understand how economic shocks shape marriage outcomes, I examine spousal characteristics among individuals who marry after the event. This analysis allows me to assess whether job displacement affects the quality of matches in the marriage market. If displaced individuals marry partners with systematically different characteristics, this would suggest that economic shocks influence not only whether individuals marry but also whom they marry.

The analysis is restricted to individuals who enter marriage after the event date. For these individuals, I compare spousal characteristics between the treatment and control groups, measured in the four quarters preceding the marriage quarter. The regressions control for pre-displacement individual and establishment characteristics, including age, industry, establishment size, tenure, and pre-displacement earnings.

Table 5 reports the results. Overall, job displacement is associated with limited changes in spouse characteristics, but the differences that do appear are concentrated in spouse education and earnings. Spousal age and the probability that the spouse has any labor income are not significantly different between the treatment and control groups for either men or women. Displaced men and women are less likely to marry spouses with a college degree or above, and their spouses have lower pre-marriage earnings than those of comparable non-displaced individuals.

For men, displaced individuals are significantly less likely to marry spouses with a college degree or above, by 4.8 percentage points relative to comparable non-displaced men. At the same time, spouses' pre-marriage earnings are lower by about 345 thousand KRW, while spousal age and the probability that the spouse has any labor income do not differ significantly between displaced and non-displaced men. A similar pattern is observed for women. Displaced women are 2.0 percentage points less likely to marry a spouse with a college degree or above, and spouses' pre-marriage earnings are lower by about 311 thousand KRW.

These results provide suggestive evidence that job displacement is associated with some changes in the observable socioeconomic characteristics of spouses among those who marry. However, the evidence does not point to broad changes in spouse characteristics, and the patterns are similar for men and women. Instead, for both genders, displaced individuals tend to marry spouses with somewhat lower educational attainment and slightly lower pre-marriage earnings than comparable non-displaced individuals. Unlike the marriage-entry effects, which differ sharply by gender, the patterns for spouse characteristics may suggest a similar weakening of displaced individuals' position in the marriage market for both men and women. By reducing perceived economic stability, job displacement may narrow the effective set of potential matches or reduce displaced individuals' ability to match with spouses with stronger socioeconomic characteristics. However, because the analysis is conditional on marriage and the differences are modest, the results should be interpreted as suggestive evidence of changes in observable spouse characteristics rather than direct evidence of a broad decline in match quality.

5 Conclusion

This paper examines the causal effect of negative economic shocks on marriage and how these effects differ by gender. Using administrative data from South Korea, I exploit job displacement due to plant closures as an exogenous source of economic shocks. While the relationship between economic conditions and marriage has been widely studied, isolating its causal effects remains empirically challenging. Existing studies have addressed this issue

by exploiting exogenous labor market shocks, but evidence remains limited on how men and women respond to job loss in the transition into marriage. This paper addresses this gap by examining marriage responses to job displacement separately for men and women. It provides insight into how economic resources may play fundamentally different roles in shaping marriage decisions for men and women.

The estimates reveal a clear gender asymmetry in marriage responses to job displacement. Over the five years following job loss, men experience an average decline of about 1 percentage point in the likelihood of being married. By contrast, women’s likelihood of being married increases after displacement, most noticeably in the year of job loss and the following year. The estimated five-year average effect is 1.7 percentage points higher for women than for men. This contrast persists when women are reweighted to match men’s pre-displacement characteristics, indicating that the gender difference is unlikely to be explained by compositional differences alone. The heterogeneity results further suggest that women’s marriage responses are stronger among those exposed to more severe or persistent economic disruptions, whereas the patterns for men are less closely tied to measured labor market losses. Among those who marry, job displacement is associated with only modest changes in observable spouse characteristics, mainly along education and pre-marriage earnings, rather than broad changes in the characteristics of spouses.

Future research could extend these findings by more directly identifying the mechanisms underlying the observed gender asymmetry. The heterogeneity analyses in this paper show that women’s marriage responses are stronger among those facing more severe or persistent economic disruptions, whereas men’s responses are less consistently related to measured labor market losses. However, these patterns do not provide direct evidence on the channels through which job displacement affects marriage entry and how these channels operate differently by gender. Further work could examine whether these responses are driven by current income losses, changes in employment stability, changes in expected future economic prospects, or other non-economic factors. Such evidence would provide a deeper understanding of how negative economic shocks translate into marriage entry for men and women.

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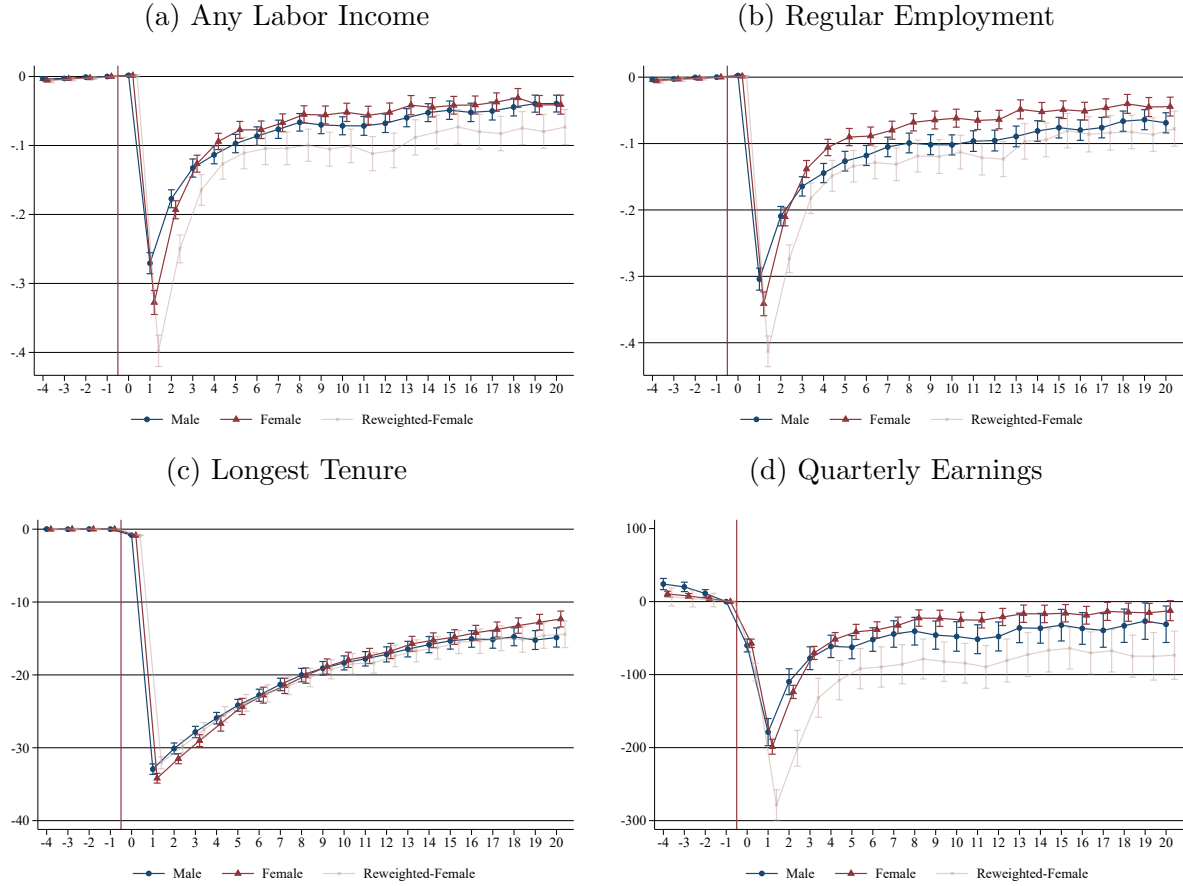
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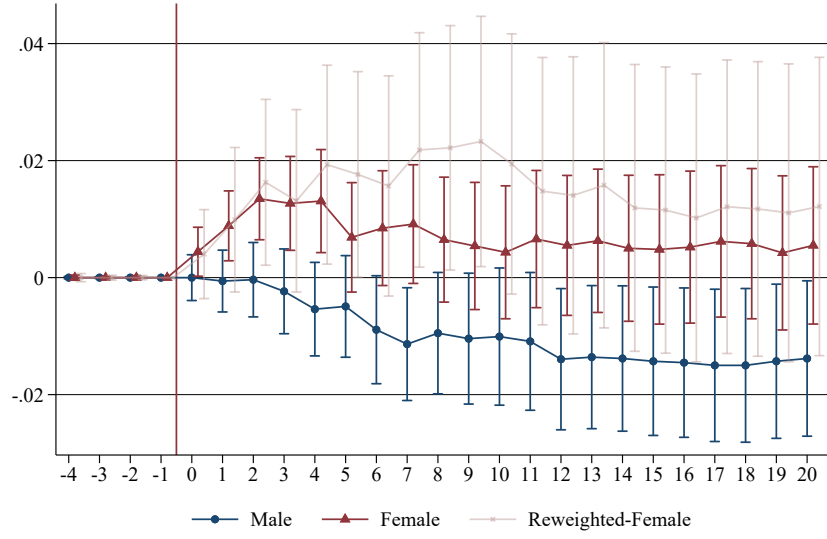
Figures

Figure 1: Effects of Job Displacement on Labor Market Outcomes



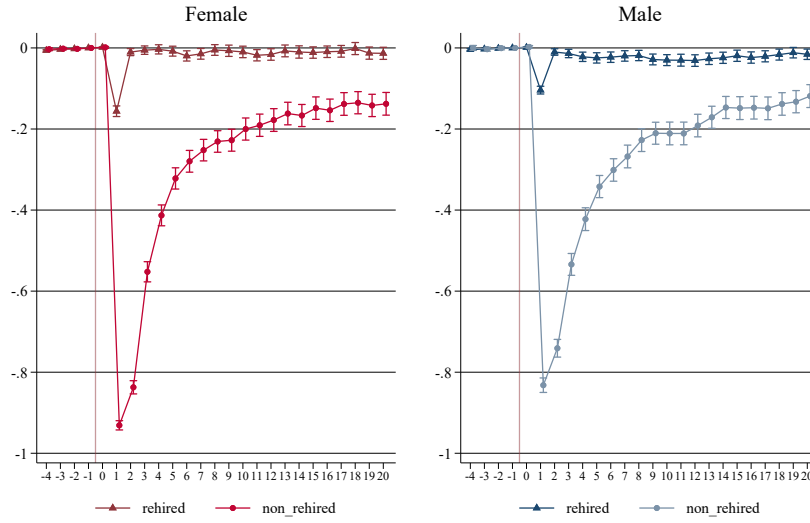
Notes: The figures present dynamic difference-in-differences estimates of the impact of job displacement on employment (including all employment statuses) (Panel (a)), regular employment (Panel (b)), length of the longest job spell (Panel (c)), and quarterly earnings of wage workers (Panel (d)). Earnings in Panel (d) are reported in units of 10,000 KRW. The analysis sample consists of individuals born between 1983 and 1995 who are continuously observed in the panel. The specification includes calendar time and individual fixed effects. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure 2: Effects of Job Displacement on Marriage

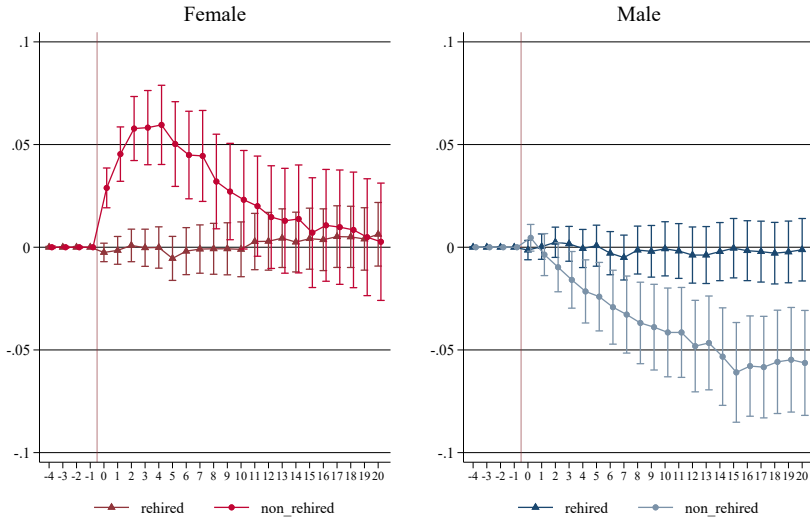


Notes: The figure presents dynamic difference-in-differences estimates of the impact of job displacement on marriage. The sample and fixed effects are identical to those in Figure 1. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure 3: Effect of Job Displacement on Marriage: Heterogeneity by Re-employment Status



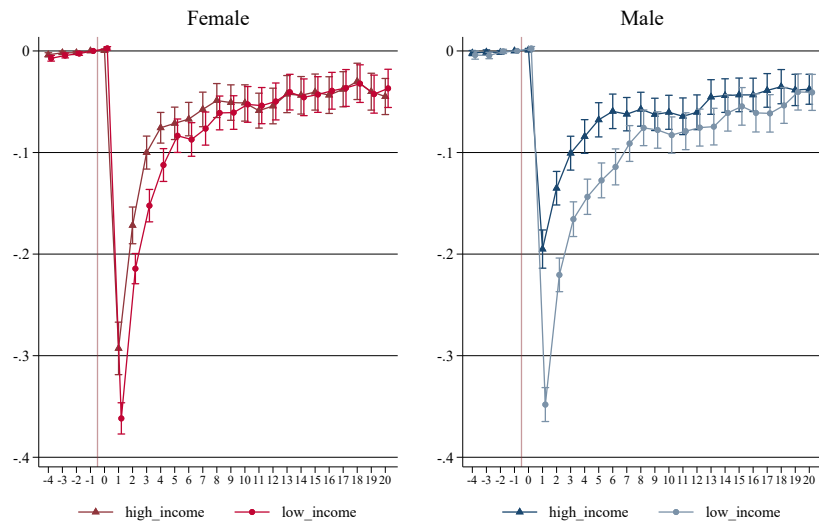
(a) Any Labor Income



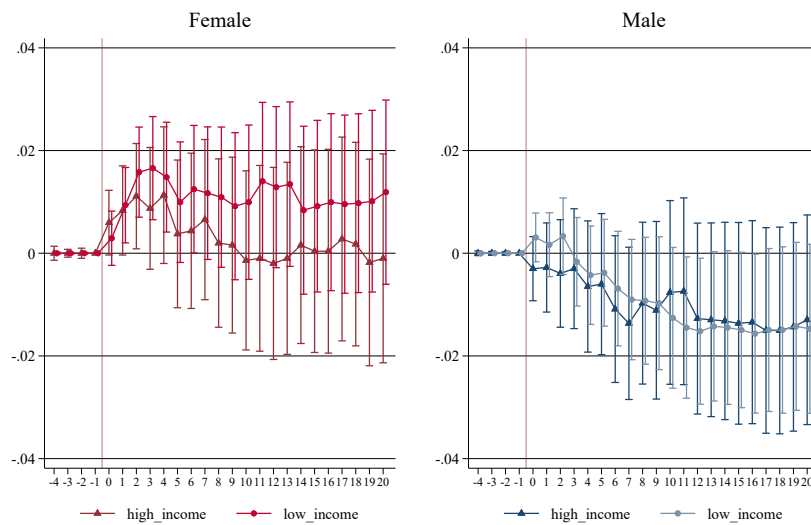
(b) Probability of Being Married

Notes: This figures present heterogeneous effects of job displacement on employment and marriage by re-employment status, where re-employment is defined as obtaining regular employment within two quarters of displacement. The sample and fixed effects are identical to those in Figure 1. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure 4: Effect of Job Displacement on Marriage: Heterogeneity by Pre-displacement Earnings



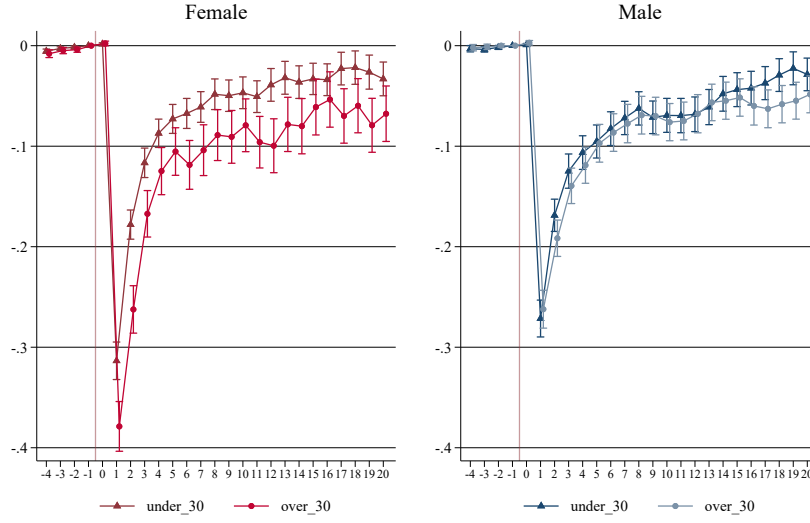
(a) Any Labor Income



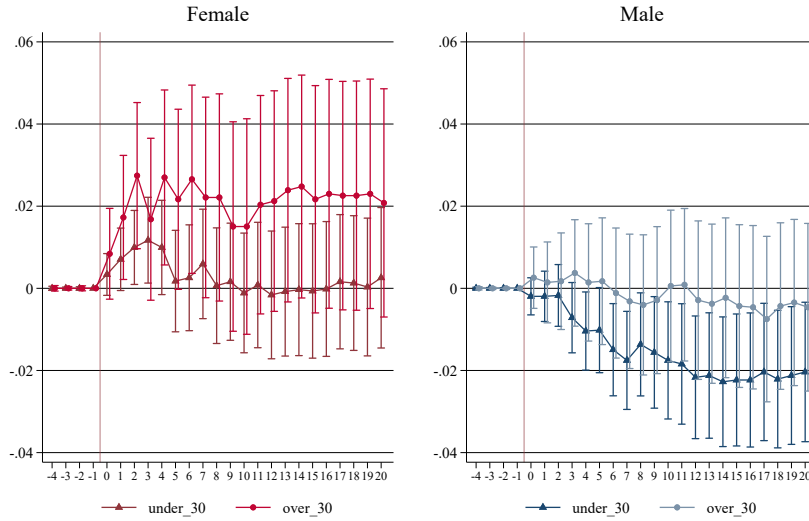
(b) Probability of Being Married

Notes: This figure presents heterogeneous effects of job displacement on employment and marriage by pre-displacement earnings (bottom 50% vs. top 50%). The sample and fixed effects are identical to those in Figure 1. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure 5: Effect of Job Displacement on Marriage: Heterogeneity by Age



(a) Any Labor Income



(b) Probability of Being Married

Notes: This figure presents heterogeneous effects of job displacement on employment and marriage by age (below 30 vs. 30 and above). The sample and fixed effects are identical to those in Figure 1. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Tables

Table 1: Descriptive Statistics for Displaced and Non-Displaced Workers

	(1) Displaced Women	(2) Non-Displaced Women	SMD	(3) Displaced Men	(4) Non-Displaced Men	SMD
Panel A: Individual Characteristics						
Age ($c-1$)	26.801 (3.235)	26.807 (3.228)	0.0018	28.489 (3.134)	28.497 (3.110)	0.0026
Current tenure ($c-1$)	30.974 (19.135)	30.875 (19.103)	-0.0052	29.958 (17.426)	29.983 (17.296)	0.0014
Total quarterly earnings ($c-4$)	551.344 (162.776)	553.497 (163.308)	0.0132	711.609 (248.108)	712.309 (247.019)	0.0028
Panel B: Establishment Characteristics						
Revenue ($c-4$)	2,840.122 (4,482.752)	2,946.860 (4,509.614)	0.0237	3,392.503 (4,337.409)	3,805.772 (5,209.882)	0.0862
Revenue ($c-8$)	2,982.758 (4,833.387)	2,941.305 (4,765.105)	-0.0086	3,021.206 (3,867.586)	3,474.558 (5,289.854)	0.0978
Employee 5-9	0.425 (0.494)	0.425 (0.494)	0	0.330 (0.470)	0.330 (0.470)	0
Employee 10-29	0.333 (0.471)	0.333 (0.471)	0	0.273 (0.446)	0.273 (0.446)	0
Employee 30-49	0.073 (0.261)	0.073 (0.261)	0	0.087 (0.282)	0.087 (0.282)	0
Employee 50-99	0.079 (0.270)	0.079 (0.270)	0	0.196 (0.397)	0.196 (0.397)	0
Employee 100-199	0.039 (0.194)	0.039 (0.194)	0	0.089 (0.285)	0.089 (0.285)	0
Employee 200-299	0.005 (0.069)	0.005 (0.069)	0	0.011 (0.105)	0.011 (0.105)	0
Employee 300-999	0.046 (0.209)	0.046 (0.209)	0	0.013 (0.113)	0.013 (0.113)	0
Number of Observations	10,165	10,165		8,536	8,536	

Notes: This table presents the pre-displacement characteristics of displaced and non-displaced workers, by gender. Age and tenure are reported as observed at quarter $t = c - 1$, while the remaining firm-level variables are based on observations at $t = c - 4$. Earnings and revenue variables have been adjusted for inflation and converted from nominal to real terms. Quarterly earnings are expressed in units of 10,000 KRW, and firm revenue in units of 1 million KRW. Standard deviations are in brackets.

Table 2: Descriptive Statistics of Displaced Workers in the Reference Quarter

	(1) Baseline Sample Women	(2) Reweighted Women	(3) Baseline Sample Men
Panel A: Individual Characteristics			
Age ($c-1$)	26.801 [3.235]	28.328 [3.222]	28.489 [3.134]
Current tenure ($c-1$)	30.974 [19.135]	28.519 [17.190]	29.958 [17.426]
Total quarterly earnings ($c-4$)	551.344 [162.776]	632.292 [181.928]	711.609 [248.108]
Panel B: Establishment Characteristics			
Revenue ($c-4$)	2,840.122 [4,482.752]	3,119.095 [3,986.056]	3,392.503 [4,337.409]
Revenue ($c-8$)	2,982.758 [4,833.387]	3,085.704 [4,526.179]	3,021.206 [3,867.586]
Employee 5-9	0.425 [0.494]	0.395 [0.489]	0.330 [0.470]
Employee 10-29	0.333 [0.471]	0.291 [0.454]	0.273 [0.446]
Employee 30-49	0.073 [0.261]	0.091 [0.288]	0.087 [0.282]
Employee 50-99	0.079 [0.270]	0.131 [0.337]	0.196 [0.397]
Employee 100-199	0.039 [0.194]	0.073 [0.260]	0.089 [0.285]
Employee 200-299	0.005 [0.069]	0.007 [0.082]	0.011 [0.105]
Employee 300-999	0.046 [0.209]	0.013 [0.082]	0.013 [0.113]
Number of Observations	10,165	10,165	8,536

Notes: This table presents the pre-displacement characteristics of displaced workers, by gender. Age, tenure, and quarterly earnings are observed at time $t = c - 1$, while revenue and establishment-level variables are measured at $t = c - 4$. Earnings and revenue are deflated to real terms. Column (2) contains displaced women reweighted to match men. Quarterly earnings are expressed in units of 10,000 KRW, and firm revenue in units of 1 million KRW. Standard deviations are in brackets.

Table 3: Aggregate Effects of Job Displacement on Marriage and Labor Market Outcomes

	(1) Probability of Being Married	(2) Any Labor Income	(3) Regular Employment	(4) Longest Tenure	(5) Quarterly Earnings
Panel A. Male					
Treated $\times$ 1(Year of displacement)	-0.0008 (0.0026)	-0.1430*** (0.0047)	-0.1670*** (0.0053)	-22.94*** (0.2780)	-120.40*** (6.5540)
Treated $\times$ 1(After 1year)	-0.0076* (0.0043)	-0.0917*** (0.0059)	-0.1220*** (0.0068)	-23.55*** (0.4020)	-68.86*** (7.1350)
Treated $\times$ 1(After 2year)	-0.0102* (0.0056)	-0.0682*** (0.0058)	-0.0981*** (0.0070)	-18.82*** (0.4710)	-60.27*** (8.8560)
Treated $\times$ 1(After 3year)	-0.0139** (0.0062)	-0.0555*** (0.0059)	-0.0838*** (0.0073)	-16.21*** (0.5410)	-51.99*** (9.5860)
Treated $\times$ 1(After 4year)	-0.0147** (0.0065)	-0.0447*** (0.0058)	-0.0699*** (0.0072)	-15.05*** (0.5900)	-47.85*** (11.1500)
Observations	409,728	409,728	409,728	409,728	409,728
Panel B. Female					
Treated $\times$ 1(Year of displacement)	0.0099*** (0.0029)	-0.1590*** (0.0048)	-0.1690*** (0.0051)	-23.88*** (0.2620)	-118.00*** (3.4340)
Treated $\times$ 1(After 1year)	0.0094** (0.0046)	-0.0763*** (0.0056)	-0.0884*** (0.0059)	-23.82*** (0.5330)	-46.60*** (4.6390)
Treated $\times$ 1(After 2year)	0.0057 (0.0056)	-0.0520*** (0.0059)	-0.0622*** (0.0062)	-18.56*** (0.5140)	-29.52*** (4.8420)
Treated $\times$ 1(After 3year)	0.0054 (0.0062)	-0.0423*** (0.0062)	-0.0507*** (0.0065)	-15.62*** (0.5120)	-23.15*** (5.5540)
Treated $\times$ 1(After 4year)	0.0054 (0.0065)	-0.0350*** (0.0062)	-0.0430*** (0.0065)	-13.49*** (0.5250)	-21.04*** (5.8970)
Observations	487,920	487,920	487,920	487,920	487,920
Constant	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes	Yes

Notes: This table presents estimated effects of job displacement on marriage entry and labor market outcomes. “Year of displacement” refers to the four quarters including the quarter of displacement. “After year” refers to the four quarters following displacement (i.e., 1, 2, and 3 years after displacement). All models include calendar fixed effects and individual fixed effects. Standard errors clustered at the company level are reported in parentheses.

Table 4: Gender Differences in Job Displacement Effects

	(1) Probability of Being Married	(2) Any Labor Income	(3) Regular Employment	(4) Longest Tenure	(5) Quarterly Earnings
Panel A. Baseline Female vs Male					
Post $\times$ Treat	-0.010*** (0.004)	-0.079*** (0.005)	-0.106*** (0.006)	-19.100*** (0.403)	-68.680*** (7.385)
Post $\times$ Treat $\times$ Female	0.017** (0.007)	0.007 (0.006)	0.025*** (0.007)	0.346 (0.540)	22.430*** (8.387)
Panel B. Reweighted Female vs Male					
Post $\times$ Treat	-0.010** (0.005)	-0.079*** (0.005)	-0.106*** (0.003)	-19.100*** (0.403)	-68.680*** (7.368)
Post $\times$ Treat $\times$ Female	0.024** (0.011)	-0.033*** (0.009)	-0.020*** (0.010)	0.036 (0.643)	-32.660*** (12.620)
Constant	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes	Yes
Observations	935,050	935,050	935,050	935,050	935,050

Notes: This table presents estimated effects of job displacement on marriage entry and labor market outcomes. Post is an indicator for all periods after job displacement. All models include calendar fixed effects and individual fixed effects. Standard errors clustered at the company level are reported in parentheses.

Table 5: Comparison of Spouse Characteristics

	(1) Spouse Age	(2) Spouse College or Above	(3) Spouse Total Earnings	(4) Spouse Earning Ratio	(5) Spouse Ever Work
PANEL A. Male					
Treat	0.090 (0.099)	-0.048*** (0.014)	-34.47** (14.030)	0.008 (0.008)	-0.011 (0.013)
Observations	3,903	3,903	3,903	3,818	3,903
PANEL B. Female					
Treat	0.016 (0.075)	-0.020* (0.011)	-31.11** (15.950)	0.013* (0.007)	-0.010 (0.010)
Observations	6,568	6,568	6,568	6,420	6,568
Constant	Yes	Yes	Yes	Yes	Yes
Control	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes	Yes

Notes: This table presents regression estimates of the characteristics of spouses of workers who enter marriage, comparing outcomes between the treatment and control groups. The dependent variables are listed in column headers. All variables are measured four quarters prior to marriage. All regressions control for pre-displacement characteristics and calendar-time fixed effects. Spouse earnings are reported in units of 10,000 KRW. Robust standard errors in parentheses. Standard errors are clustered at the company level.

Appendix

A. Details on Matching Procedure

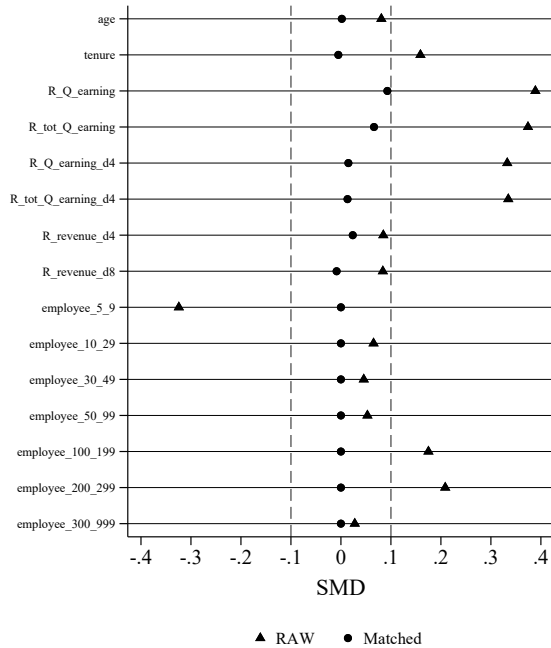
Matching is implemented using coarsened exact matching (CEM) without replacement. The procedure is conducted separately for men and women.

Age at $t = c - 1$ is grouped into five bins: 20–23, 24–26, 27–29, 30–32, and 33–35. Establishment size at $t = c - 4$ is classified into nine categories based on the Employer–Employee Data. Industry at $t = c - 4$ is coded into 21 categories following the Korean Standard Industrial Classification. Tenure at $t = c - 1$ is measured in months and grouped into one-year intervals.

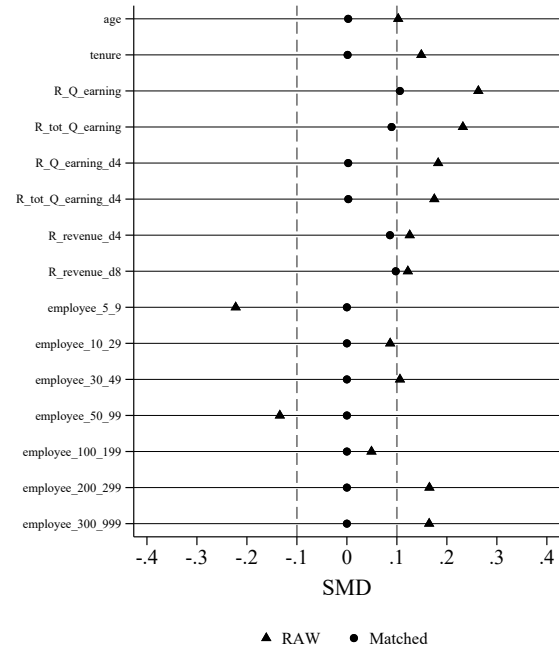
In addition, observations in the top and bottom 1% of the revenue distribution and the top 1% of the quarterly earnings distribution are excluded separately for men and women to mitigate the influence of extreme values.

B. Additional Figures and Tables

Figure B.1: Standardized Mean Differences Before and After Matching



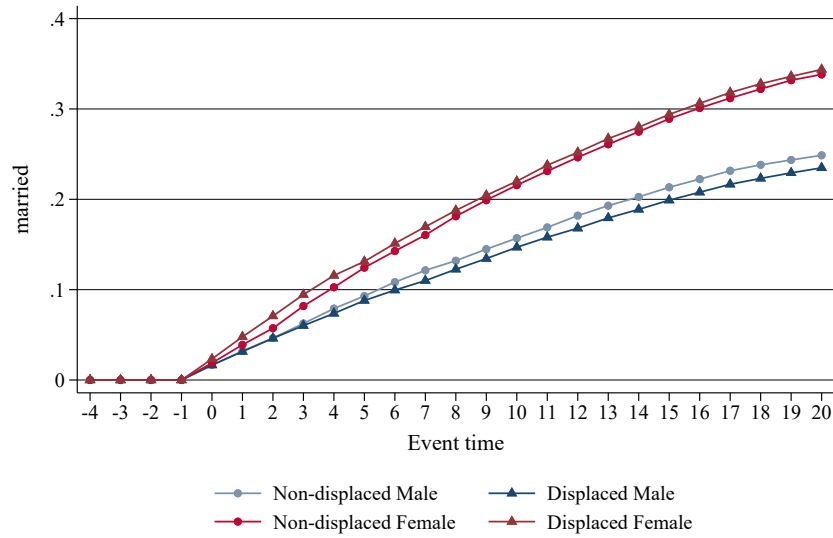
(a) Female Sample



(b) Male Sample

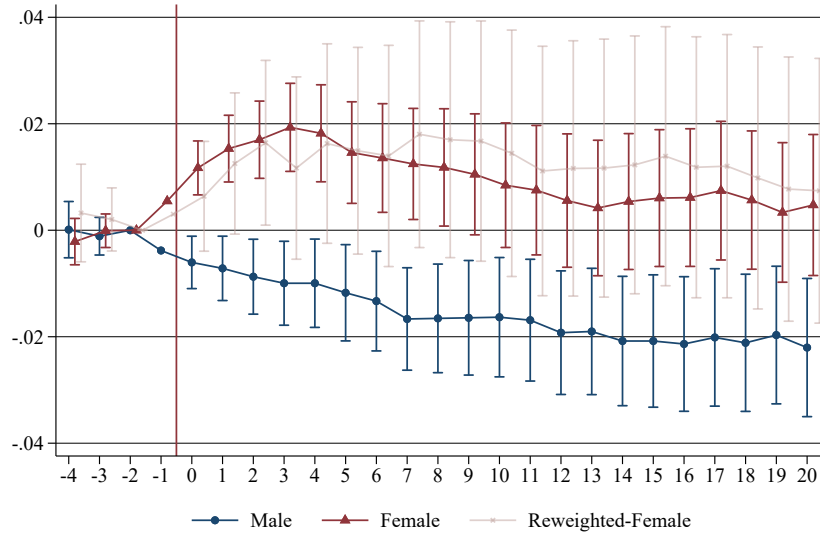
Notes: This figure presents standardized mean differences (SMDs) for covariates before and after matching. Black triangles indicate the raw sample, and black circles indicate the matched sample. The dashed vertical lines correspond to ± 0.1 , a commonly used threshold for acceptable balance.

Figure B.2: Mean Trends in the Probability of Being Married



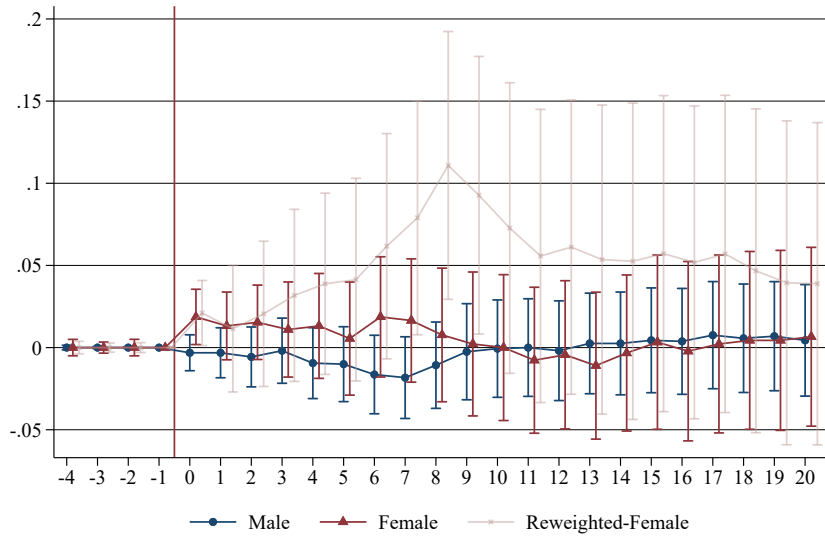
Notes: This figure plots mean trends in the probability of being married by gender and displacement status. Triangles indicate displaced workers, and circles indicate non-displaced workers. Red-toned lines correspond to women, while blue-toned lines correspond to men. The sample is restricted to individuals who are unmarried during the four quarters prior to displacement but may marry thereafter.

Figure B.3: Effects of Job Displacement on Marriage



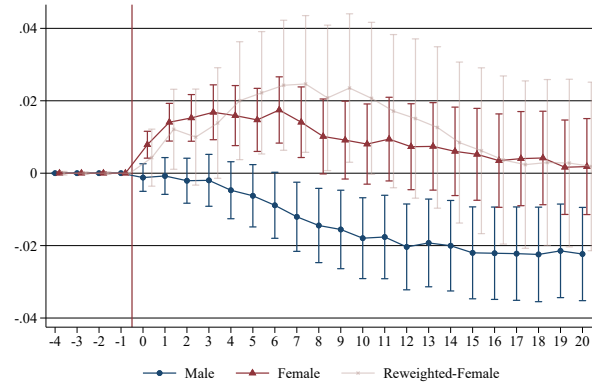
Notes: The figure presents dynamic difference-in-differences estimates of the impact of job displacement on marriage. The sample and fixed effects are identical to those in Figure 1, except that the sample is restricted to individuals who are unmarried up to four quarters prior to displacement but may marry thereafter, in order to examine whether marriage patterns between the treatment and control groups differ prior to the event. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure B.4: Effects of Job Displacement on Marriage: 50+ Employees, 2+ Years Tenure

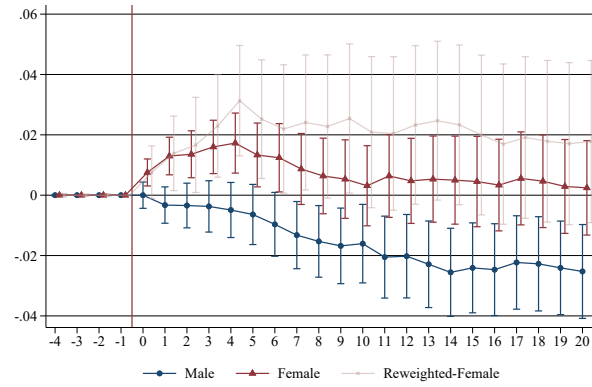


Notes: The figure present dynamic difference-in-differences estimates of the impact of job displacement on marriage. The sample and fixed effects are identical to those in Figure 1, except that the sample is restricted to workers employed at establishments with at least 50 employees four quarters prior to displacement and with at least two years of tenure one quarter prior to displacement. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

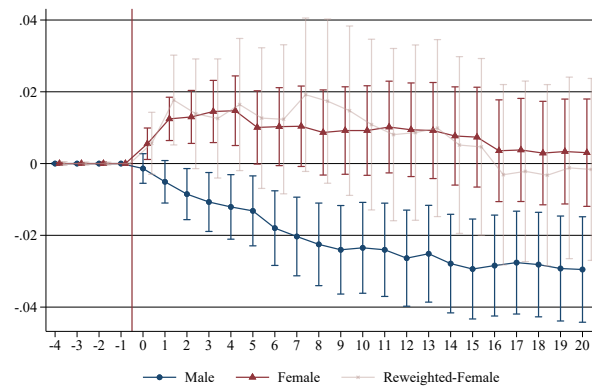
Figure B.5: Robustness Check: Effects of Job Displacement on Marriage



(a) Excluding revenue variables



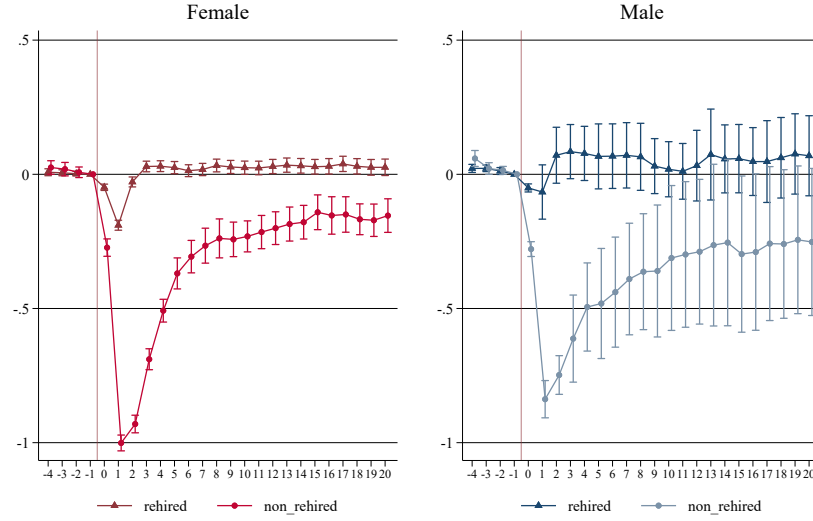
(b) Including revenue variables: Once-treated vs. never-treated



(c) Excluding revenue variables: Once-treated vs. never-treated

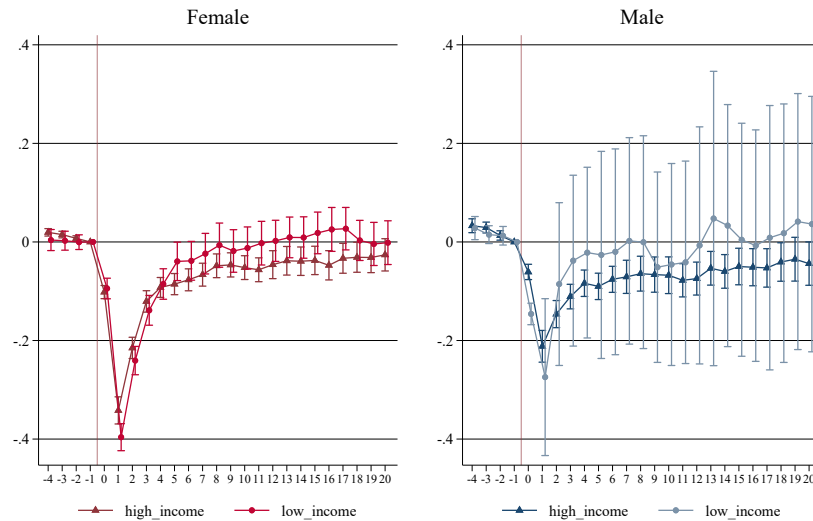
Notes: The figures present dynamic difference-in-differences estimates of the impact of job displacement on marriage. The sample and fixed effects are identical to those in Figure 1. Panel (a) presents results using a matched sample that excludes revenue variables. Panel (b) uses a matched sample that includes revenue variables and restricts the sample to once-treated individuals and never-treated controls. Panel (c) excludes revenue variables in matching while restricting the sample to once-treated individuals and never-treated controls. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure B.6: Effects of Job Displacement on Relative Earnings: Heterogeneity by Re-employment Status



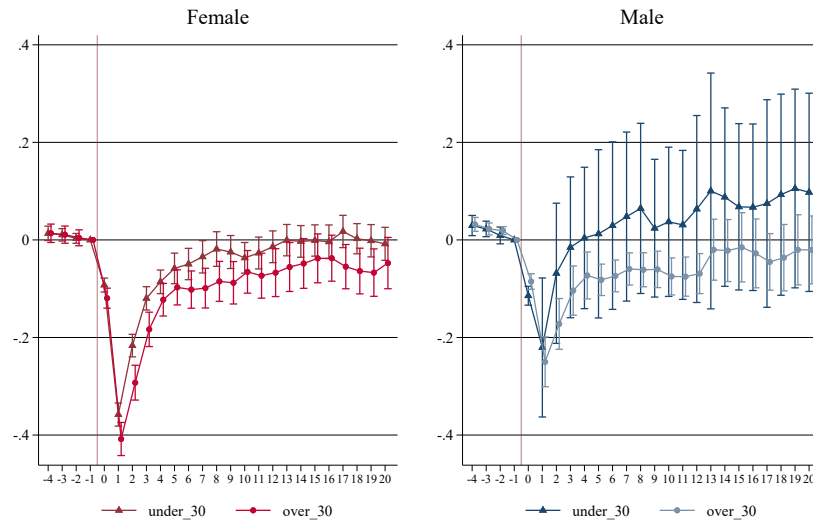
Notes: This figure presents heterogeneous effects of job displacement on relative earnings by re-employment status, where re-employment is defined as obtaining regular employment within two quarters of displacement. The sample and fixed effects are identical to those in Figure 1. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure B.7: Effects of Job Displacement on Relative Earnings: Heterogeneity by Pre-displacement Earnings



Notes: This figure presents heterogeneous effects of job displacement on relative earnings by pre-displacement earnings (bottom 50% vs. top 50%). The sample and fixed effects are identical to those in Figure 1. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Figure B.8: Effects of Job Displacement on Relative Earnings: Heterogeneity by Age



Notes: This figure presents heterogeneous effects of job displacement on relative earnings by age (below 30 vs. 30 and above). The sample and fixed effects are identical to those in Figure 1. Each dot represents the interaction coefficient between treatment status and event time, and capped lines indicate 95% confidence intervals. Standard errors are clustered at the company level.

Table B.1: Descriptive Characteristics of Single- and Multi-Establishment Firms

	Multi-Company Firms	Single-Company Firms
A. Coverage in 2016		
Share of firms	0.02	0.98
Share of workers	0.43	0.57
B. Characteristics		
Average number of workers	127	6
Age	43	49
Annual income	2,373	1,693
Tenure	40	67
Regular worker share	0.57	0.18
Daily worker share	0.16	0.08
Full-time worker share	0.41	0.10
Married share	0.77	0.82
Married share(83-95)	0.21	0.24
Revenue	22,176	401
C. Industry Shares		
Manufacturing (C)	0.22	0.09
Wholesale and retail trade (G)	0.10	0.25
Transportation and storage (H)	0.03	0.11
Accommodation and food service (I)	0.05	0.15
Real estate activities (L)	0.05	0.11

Notes: This table compares descriptive characteristics of single-establishment and multi-establishment firms in 2016. Single-establishment firms refer to firms with one establishment, while multi-establishment firms refer to firms with more than one establishment. In Panel A, the firm share is calculated using firms as the unit of observation, while the worker share is calculated using workers as the unit of observation. Panels B and C use the firm as the unit of observation. Worker characteristics in Panel B are measured as firm-level averages or shares among workers within each firm. Annual income is expressed in units of 10,000 KRW, and revenue is expressed in units of 1 million KRW.

Table B.2: Descriptive Statistics for Displaced and Non-Displaced Workers

	(1) Displaced Women	(2) Non-Displaced Women	SMD	(3) Displaced Men	(4) Non-Displaced Men	SMD
A	0.000 (0.010)	0.000 (0.010)	0	0.001 (0.024)	0.001 (0.024)	0
B	0.000 (0.000)	0.000 (0.000)	0	0.000 (0.000)	0.000 (0.000)	0
C	0.080 (0.272)	0.080 (0.272)	0	0.397 (0.489)	0.397 (0.489)	0
D	0.000 (0.000)	0.000 (0.000)	0	0.000 (0.000)	0.000 (0.000)	0
E	0.000 (0.000)	0.000 (0.000)	0	0.000 (0.019)	0.000 (0.019)	0
F	0.018 (0.132)	0.018 (0.132)	0	0.043 (0.202)	0.043 (0.202)	0
G	0.082 (0.275)	0.082 (0.275)	0	0.091 (0.288)	0.091 (0.288)	0
H	0.006 (0.080)	0.006 (0.080)	0	0.016 (0.124)	0.016 (0.124)	0
I	0.087 (0.023)	0.087 (0.023)	0	0.162 (0.369)	0.162 (0.369)	0
J	0.023 (0.149)	0.023 (0.149)	0	0.035 (0.183)	0.035 (0.183)	0
K	0.001 (0.024)	0.001 (0.024)	0	0.001 (0.024)	0.001 (0.024)	0
L	0.002 (0.046)	0.002 (0.046)	0	0.003 (0.055)	0.003 (0.055)	0
M	0.059 (0.236)	0.059 (0.236)	0	0.043 (0.202)	0.043 (0.202)	0
N	0.091 (0.288)	0.091 (0.288)	0	0.110 (0.313)	0.110 (0.313)	0
O	0.000 (0.000)	0.000 (0.000)	0	0.000 (0.000)	0.000 (0.000)	0
P	0.046 (0.209)	0.046 (0.209)	0	0.008 (0.088)	0.008 (0.088)	0
Q	0.467 (0.499)	0.467 (0.499)	0	0.055 (0.229)	0.055 (0.229)	0
R	0.006 (0.077)	0.006 (0.077)	0	0.009 (0.094)	0.009 (0.094)	0
S	0.032 (0.176)	0.032 (0.176)	0	0.026 (0.160)	0.026 (0.160)	0
Number of Observations	10,165	10,165		8,536	8,536	

Notes: This table presents the pre-displacement characteristics of displaced and non-displaced workers, by gender. In addition to Table 1, this table includes industry classification variables in the descriptive statistics. Standard deviations are in brackets.

Table B.3: Descriptive Statistics of Displaced Workers in the Reference Quarter

	(1) Baseline Sample Women	(2) Reweighted Women	(3) Baseline Sample Men
A	0.000 [0.010]	0.000 [0.011]	0.001 [0.024]
B	0.000 [0.000]	0.000 [0.000]	0.000 [0.000]
C	0.080 [0.272]	0.309 [0.462]	0.397 [0.489]
D	0.000 [0.000]	0.000 [0.000]	0.000 [0.000]
E	0.000 [0.000]	0.000 [0.000]	0.000 [0.019]
F	0.018 [0.132]	0.045 [0.207]	0.043 [0.202]
G	0.082 [0.275]	0.114 [0.318]	0.091 [0.288]
H	0.006 [0.080]	0.014 [0.117]	0.016 [0.124]
I	0.087 [0.023]	0.198 [0.399]	0.162 [0.369]
J	0.023 [0.149]	0.042 [0.200]	0.035 [0.183]
K	0.001 [0.024]	0.000 [0.021]	0.001 [0.024]
L	0.002 [0.046]	0.003 [0.052]	0.003 [0.055]
M	0.059 [0.236]	0.050 [0.218]	0.043 [0.202]
N	0.091 [0.288]	0.114 [0.318]	0.110 [0.313]
O	0.000 [0.000]	0.000 [0.000]	0.000 [0.000]
P	0.046 [0.209]	0.010 [0.100]	0.008 [0.088]
Q	0.467 [0.499]	0.064 [0.245]	0.055 [0.229]
R	0.006 [0.077]	0.010 [0.100]	0.009 [0.094]
S	0.032 [0.176]	0.026 [0.160]	0.026 [0.160]
Number of Observations	10,165	10,165	8,536

Notes: This table presents the pre-displacement characteristics of displaced workers, by gender. In addition to Table 2, this table includes industry classification variables in the descriptive statistics. Column (2) contains displaced women reweighted to match men. Standard deviations are in brackets.

Table B.4: Effects of Job Displacement on Marriage and Labor Market Outcomes: Reweighted Female Sample

	(1) Probability of Being Married	(2) Any Labor Income	(3) Regular Employment	(4) Longest Tenure	(5) Quarterly Earnings
Treated $\times$ 1(Year of displacement)	0.0108* (0.0056)	-0.2000*** (0.0070)	-0.2140*** (0.0073)	-22.56*** (0.3230)	-180.10*** (8.0840)
Treated $\times$ 1(After 1year)	0.0186** (0.0089)	-0.1090*** (0.0102)	-0.1330*** (0.0109)	-23.32*** (0.5620)	-96.85*** (12.4000)
Treated $\times$ 1(After 2year)	0.0199* (0.0108)	-0.1010*** (0.0109)	-0.1160*** (0.0116)	-19.10*** (0.6540)	-86.80*** (12.9100)
Treated $\times$ 1(After 3year)	0.0133 (0.0122)	-0.0843*** (0.0112)	-0.0961*** (0.0121)	-16.47*** (0.7550)	-74.13*** (13.6600)
Treated $\times$ 1(After 4year)	0.0113 (0.0127)	-0.0764*** (0.0113)	-0.0821*** (0.0123)	-15.02*** (0.8480)	-74.99*** (13.9600)
Constant	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes	Yes
Observations	487,920	487,920	487,920	487,920	487,920

Notes: This table presents the estimated effects of job displacement on marriage entry and labor market outcomes for the reweighted female sample. “Year of displacement” refers to the four quarters including the quarter of displacement. “After year” refers to the four quarters following displacement (i.e., 1, 2, and 3 years after displacement). All models include calendar fixed effects and individual fixed effects. Standard errors clustered at the company level are reported in parentheses.

Table B.5: Effects of Job Displacement on Marriage and Labor Market Outcomes

	(1) Probability of Being Married	(2) Any Labor Income	(3) Regular Employment	(4) Longest Tenure	(5) Quarterly Earnings
Panel A. Male					
Treated $\times$ 1(Post)	-0.0097** (0.0047)	-0.0786*** (0.0046)	-0.1060*** (0.0057)	-19.10*** (0.40)	-68.68*** (7.34)
Observations	426,800	426,800	426,800	426,800	426,800
Panel B. Female					
Treated $\times$ 1(Post)	0.0070 (0.0046)	-0.0712*** (0.0048)	-0.0808*** (0.0051)	-18.75*** (0.41)	-46.25*** (3.91)
Observations	508,250	508,250	508,250	508,250	508,250
Panel C. Reweighted Female					
Treated $\times$ 1(Post)	0.0147 (0.0091)	-0.1120*** (0.0079)	-0.1260*** (0.0087)	-19.06*** (0.56)	-101.30*** (10.71)
Observations	508,250	508,250	508,250	508,250	508,250
Constant	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes	Yes

Notes: This table presents estimated effects of job displacement on marriage entry and labor market outcomes. Post is an indicator for all periods after job displacement. All models include calendar fixed effects and individual fixed effects. Standard errors clustered at the company level are reported in parentheses.

Table B.6: Comparison of Spouse Characteristics: Reweighted Female

	(1) Spouse Age	(2) Spouse College or Above	(3) Spouse Total Earnings	(4) Spouse Earning Ratio	(5) Spouse Ever Work
PANEL C. Reweighted Female					
Treat	0.067 (0.154)	-0.050** (0.022)	-59.04** (26.940)	0.011 (0.013)	-0.013 (0.017)
Observations	6,568	6,568	6,568	6,420	6,568
Constant	Yes	Yes	Yes	Yes	Yes
Control	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Company Cluster	Yes	Yes	Yes	Yes	Yes

Notes: This table presents regression estimates of the characteristics of spouses of reweighted female workers who enter marriage, comparing outcomes between the treatment and control groups. The dependent variables are listed in column headers. All variables are measured four quarters prior to marriage. All regressions control for pre-displacement characteristics and calendar-time fixed effects. Spouse earnings are reported in units of 10,000 KRW. Standard errors clustered at the company level are reported in parentheses.